

Name:

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Homework Assignment 0

ECE 9433, Fall 2026

Useful Equations

Wire resistance: $R = \frac{\rho L}{Wt}$

Shockley transistor model:

$$I_{ds} = \begin{cases} 0, & V_{gs} < V_t \text{ Cutoff} \\ \beta(V_{GT} - V_{ds}/2) * V_{ds}, & V_{gs} > V_t \text{ and } V_{ds} < V_{dsat} \text{ Linear} \\ \beta V_{GT}^2 / 2, & V_{ds} > V_{dsat} \text{ Saturation} \end{cases}$$

Where:

- $C_{ox} = \epsilon_{ox} / t_{ox}$
- $\beta = (\epsilon_{ox} * \mu / t_{ox}) * (W/L) = C_{ox} * \mu * W/L$
- $V_{GT} = V_{gs} - V_t$
- $V_{dsat} = V_{GT}$

For other equations, please see lecture slides.

Problem 1

Circle the most correct relationship for each of the following on the Hardware/Software Stack.

As abstraction increases, flexibility: increases decreases

As abstraction increases, productivity: increases decreases

As abstraction increases, efficiency: increases decreases

Problem 2

Circle the best answer.

Parasitic capacitances in MOSFETs are strongly influenced by:

A) Differences in doping concentration

B) Type of dopant atoms

C) Width of the transistor

D) All of the above

Problem 3

Circle True or False for each statement, whichever is the most correct.

Making a wire longer will increase its resistance:

True False

Wider wires increase capacitance, and therefore have longer delay:

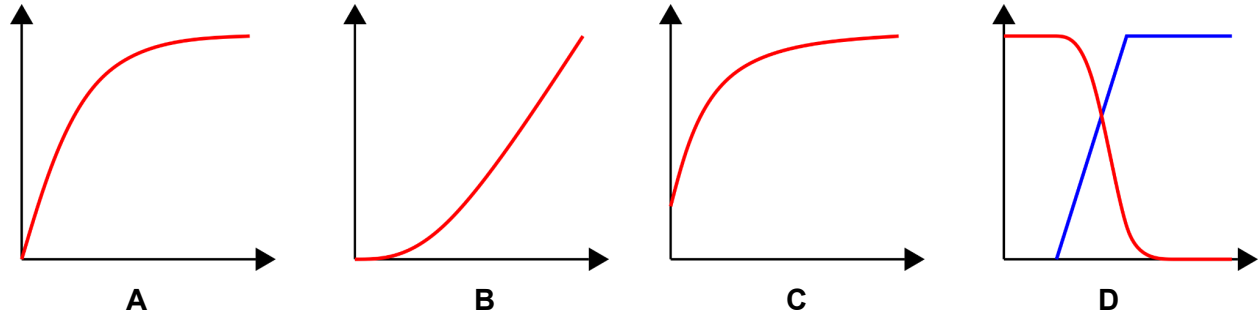
True False

Crosstalk can make the wire delay faster:

True False

Problem 4

The following graphs model behaviors of an nMOS transistor. Fill in the correct labels for the x- and y-axes on each graph



A: x-axis: V_{ds} y-axis: I_d

B: x-axis: V_{gs} y-axis: I_d

C: x-axis: V_{gs} y-axis: $\log(I_d)$

D: x-axis: Time y-axis: V

Problem 5

Why is CMOS logic generally preferred over other types of logic? Answer in 1-3 sentences.

Well balanced between power, area and noise

Problem 6

Can clock skew be helpful for circuit timing? Explain why or why not. Answer in 1-3 sentences.

Yes, it can increase the setup time but at the cost of also increasing hold time

Problem 7

A circuit has the following properties:

Clock-to-q delay: 10 ps

Propagation delay: 50 ps

Contamination delay: 45 ps

Hold time: -2 ps

Setup time: 3 ps

What is the maximum absolute clock skew, $|\phi|$, before the circuit will fail hold time? ($||$ denotes absolute value)

$$tcq + tcd \geq t_{hold} + skew$$

$$57 \geq skew$$

$$tcq + tcd - t_{hold} \geq skew$$

$$10 + 45 - (-2) \geq skew$$

Problem 8

Pipelining is best classified by which of the following microarchitectural techniques? (Circle one)

A) Caching



B) Parallelizing

C) Speculating

D) Specializing

Problem 9

A designer is trying to decide how to route a wire from Transistor X to Transistor Y, as shown in Figure 1.1. They decide that they can use two approaches. In approach A, they route from Transistor X through Via1 to Metal2, then back down to Via1 and Transistor Y. In approach B, they route from Transistor X through Via1, Metal2, and Via2 to Metal3, then back down to Transistor Y.

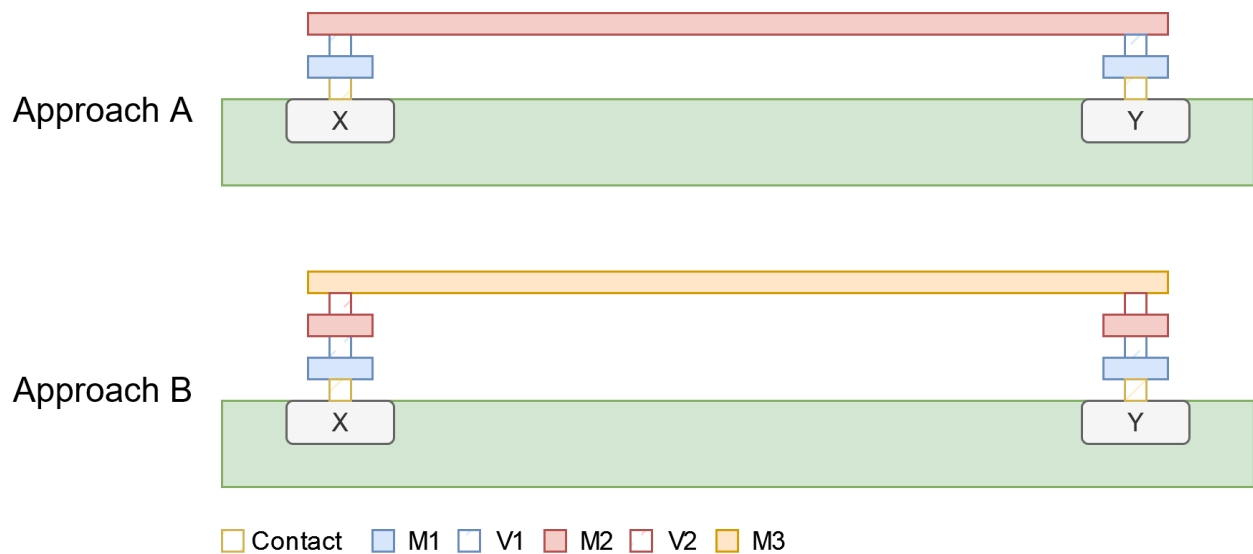


Figure 1.1

For this problem, use the following assumptions:

- Ignore contact resistance and M1 resistance
- All wires are copper with resistivity $1.68 \times 10^{-8} \Omega\text{m}$
- All vias are tungsten with resistivity $5.60 \times 10^{-8} \Omega\text{m}$
- All vias are 35 nm x 35 nm wide and 45 nm thick
- M2 is 140 nm thick and 70 nm wide
- M3 is 160 nm thick and 70 nm wide
- The horizontal length required to get from X to Y in Parts A-C is 1 μm

Be careful with unit conversions in this problem! Show your work on a separate sheet and attach to the end of this assignment.

- A) Calculate the total resistance of the route in Approach A.
- B) Calculate the total resistance of the route in Approach B. For the resistance calculation of M2, assume that each M2 connection is 35 nm x 35 nm wide and 140 nm tall. The material is still copper.

- C) Which approach should the designer use if they are trying to minimize the total resistance of the wire? Approach A
- D) The designer finds that they must move Transistor Y to a new distance L . At what value of L does it make sense to change the approach from Part C?

Problem 10

Be careful with unit conversions in this problem! Show your work on a separate sheet and attach to the end of this assignment.

- A) An NMOSFET is operating at $V_{th} = 0.7 \text{ V}$, $V_{GS} = 1.0 \text{ V}$, $V_{DS} = 0.5 \text{ V}$. Identify which operating region the transistor is in.
- B) Calculate the drain current using the Shockley model. Use $\epsilon_{ox} = 3.45 \times 10^{-11} \text{ F/m}$, $\mu_n = 200 \text{ cm}^2/\text{V}$, $t_{ox} = 1.5 \text{ nm}$, $W = 1 \text{ }\mu\text{m}$, $L = 45 \text{ nm}$.
- C) A designer wants to double the current output for the operating conditions given in Part A. Keeping all other parameters the same as Part B, calculate the required width W of the transistor.

Problem 9

Cu wire $\rho = 1.68 \times 10^{-8} \Omega \text{m}$

W via $\rho = 5.6 \times 10^{-8} \Omega \text{m}$

vias area $35 \text{ nm} \times 35 \text{ nm} \times 45 \text{ nm}$
width length thickness

M2 140 nm thick 70 nm wide

M3 160 nm thick 70 nm wide

X-Y 1 μm

L (length)

Resistance $\rho \frac{L}{A}$

A) $R_{V1} + R_{M2} + R_{V1} = R_{\text{TOT}}$

$$\rho_w \frac{th}{w \cdot l} + \rho_{cu} \frac{1 \mu\text{m}}{w \cdot th} + \rho_w \frac{th}{w \cdot l}$$

$$2 \cdot (5.6 \times 10^{-8}) \frac{45 \text{ nm}}{35 \text{ nm} \times 35 \text{ nm}} + (1.68 \times 10^{-8}) \frac{1 \mu\text{m}}{140 \text{ nm} \times 70 \text{ nm}} = 5.82 \Omega //$$

B) $R_{V1} + R_{M2} + R_{V2} + R_{M3} + R_{V2} + R_{M2} + R_{V1}$

$$2 \left[\rho_w \frac{th_{V1}}{w_{V1} \cdot l_{V1}} + \rho_{cu} \frac{th_{M2}}{w_{V1} \times l_{V1}} + \rho_w \frac{th_{V1}}{w_{V1} \cdot l_{V1}} \right] + \rho_{cu} \frac{L}{th_{M3} \cdot w_{M3}} = 13.56 \Omega //$$

D) Approach A Approach B

$$2R_{V1} + R_{M2} > 4R_{V1} + 2R'_{M2} + R_{M3}$$

$$R_{M2} > 2R_{V1} + 2R'_{M2} + R_{M3}$$

$$\rho_{cu} \frac{L}{th_{M2} \times w_{M2}} > 2R_{V1} + 2R'_{M2} + \rho_{cu} \frac{L}{th_{M3} \cdot w_{M3}}$$

$$L > \left[\rho_{cu} \frac{1}{th_{M2} \times w_{M2}} - \rho_{cu} \frac{1}{th_{M3} \times w_{M3}} \right]^{-1} \left[2R_{V1} + 2R'_{M2} \right]$$

$$L > 37.12 \mu\text{m} //$$

$$L \left[\rho_{cu} \frac{1}{th_{M2} \times w_{M2}} - \rho_{cu} \frac{1}{th_{M3} \times w_{M3}} \right] > 2R_{V1} + 2R'_{M2}$$

Problem 10

A) $V_{TH} = 0.7 \text{ V}$

$$V_{GS} = 1 \text{ V}$$

$$V_{DS} = 0.5 \text{ V}$$

$$V_{GS} > V_{TH} \rightarrow 1 \text{ V} > 0.7 \text{ V}$$

$$V_{OV} = V_{GS} - V_{TH} \rightarrow V_{OV} = 0.3 \text{ V}$$

$$V_{DS} \geq V_{GS} - V_{TH} \rightarrow 0.5 \geq 0.3 \text{ V}$$

The MOSFET is in saturation //

B) In saturation $I_{DS} = \frac{\beta V_{OV}^2}{2}$ $\beta = \left[\epsilon_{ox} * \frac{\mu}{t_{ox}} \right] \left[\frac{W}{L} \right]$

$$I_{DS} = \left[3.45 \times 10^{-11} * \frac{0.02}{1.5 \times 10^{-9}} \right] \left[\frac{1 \times 10^{-6}}{45 \times 10^{-9}} \right] \frac{0.3^2}{2} = \underline{0.46 \text{ mA} //$$

C)

$$2 I_{DS} = \frac{\beta V_{OV}^2}{2} \quad \text{or} \quad I_{DS} = \left[\epsilon_{ox} * \frac{\mu}{t_{ox}} \right] \left[\frac{W}{L} \right] \frac{V_{OV}^2}{2}$$

$$\frac{4 I_{DS}}{V_{OV}^2} = \left[\epsilon_{ox} * \frac{\mu}{t_{ox}} \right] \left[\frac{W}{L} \right]$$

$$W = \frac{L 4 I_{DS}}{V_{OV}^2} \left[\epsilon_{ox} * \frac{\mu}{t_{ox}} \right]^{-1} = \underline{2 \mu\text{m} //$$